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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations
December 2025

IS3002-1- ETHICAL HACKING AND NETWORK DEFENCE
(For ISE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Which statement best describes penetration testing?**
A) Simulating real attacks to validate the effectiveness of security controls
B) Scanning for vulnerabilities without any exploitation
C) An activity limited only to compliance auditing
D) Installing security patches automatically
2. **Which Kali Linux framework is most commonly used for exploitation?**
A) Metasploit Framework
B) Hydra
C) Aircrack-ng
D) Burp Suite
3. **Which of the following is a reputational benefit of ethical hacking for an organization?**
A) Losing customer trust
B) Demonstrating commitment to cybersecurity
C) Increasing insider threats
D) Encouraging illegal hacking
4. **Which of the following tools is commonly used for vulnerability scanning?**
A) Nmap
B) Nessus
C) Wireshark
D) GitHub
5. **At what stage are the findings of the ethical hacking process documented and presented to stakeholders?**
A) Reporting
B) Reconnaissance
C) Exploitation
D) Scanning
6. **The collaboration of Red and Blue Teams to improve security effectiveness is called:**
A) Team Hacking
B) Threat Simulation
C) Purple Teaming
D) Penetration Testing
7. **What is the first step in an ethical hacking attempt?**
A) Exploitation
B) Reporting
C) Reconnaissance
D) Covering tracks
8. **Which of the following is NOT a common threat modeling methodology?**
A) STRIDE
B) DREAD
C) PASTA
D) FTP



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9. **DNS cache poisoning aims to:**
 - A) Overload DNS servers
 - B) Insert malicious DNS records into cache
 - C) Hijack ARP responses
 - D) Block TLS traffic
10. **Which of the following is an advantage of SSO?**
 - A) Increased password fatigue
 - B) Simplified user experience
 - C) Increased risk of brute-force attacks
 - D) None of the above
11. **Which attack injects malicious scripts into web applications?**
 - A) SQL Injection
 - B) CSRF
 - C) XSS
 - D) RFI
12. **Kerberos is commonly used for**
 - A) Email encryption
 - B) Password management
 - C) Single Sign On (SSO)
 - D) Firewall configuration
13. **Which of the following is considered obsolete and insecure?**
 - A) TLS 1.2
 - B) SSL 3.0
 - C) IPsec
 - D) STARTTLS
14. **A rogue DHCP server can:**
 - A) Provide incorrect IP and gateway settings
 - B) Disable SSL connections
 - C) Poison ARP cache
 - D) Block HTTPS traffic
15. **SQL Injection attacks target**
 - A) Databases via queries
 - B) Switches via packets
 - C) DNS via spoofing
 - D) Routers via flooding
16. **The DROWN attack targets servers supporting:**
 - A) SSH
 - B) SSLv2
 - C) TLS 1.3
 - D) DNS
17. **A proxy server primarily acts as**
 - A) A direct router
 - B) An intermediary between clients and servers
 - C) An IDS system
 - D) A packet sniffer
18. **A limitation of IDS systems is that they**
 - A) Can only operate in hardware
 - B) Cannot detect zero-day attacks
 - C) Require Internet connectivity
 - D) Block traffic directly
19. **Which method is better at detecting zero-day attacks?**
 - A) Signature-based
 - B) Anomaly-based
 - C) Stateful firewall
 - D) Proxy firewall
20. **Signature-based detection is effective against**
 - A) Unknown malware
 - B) Zero-day exploits
 - C) Known attacks with specific patterns
 - D) Encrypted traffic

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Ethical Hacking. How do organizations benefit from conducting Ethical Hacking activities? Give examples.	8	L*2	1	1
	b) What steps are involved in the Life Cycle of Penetration Testing?	8	L2	2	1
2.	a) What are the key principles and objectives of Ethical Hacking?	8	L2	1	1
	b) Explain how Kali Linux supports penetration testers in performing security assessments?	8	L2	2	1

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|--|---|----|---|---|
| 3. a) What is Vulnerability Analysis and how is it different from Penetration Testing? | 8 | L2 | 1 | 1 |
| b) What are the main phases in the Typical Life Cycle of Ethical Hacking? | 8 | L2 | 2 | 1 |

Unit – II

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|---|---|----|---|---|
| 4. a) Explain SQL injection and Cross-Site Scripting (XSS) attacks with preventive measures. | 8 | L2 | 4 | 1 |
| b) Differentiate between active and passive attacks. Discuss two methods to detect and prevent passive network attacks. | 8 | L2 | 3 | 1 |
| 5. a) Discuss the DROWN and POODLE attacks on SSL/TLS protocols. | 8 | L2 | 4 | 1 |
| b) Compare IPSec, SSL 3.0, and TLS 1.2 in terms of security features and usage. | 8 | L2 | 3 | 1 |
| 6. a) Describe how different OSI layers implement security measures. | 8 | L2 | 3 | 1 |
| b) Explain ARP cache poisoning and its impact on network security. | 8 | L2 | 4 | 1 |

Unit – III

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|---|---|----|---|---|
| 7. a) Explain the any four techniques used in Network Intrusion Detection Systems (NIDS). | 8 | L2 | 5 | 1 |
| b) Discuss briefly the role of proxies, NAT, and VPN in securing networks. | 8 | L2 | 5 | 1 |
| 8. a) Compare signature-based detection with anomaly-based detection in IDS. | 8 | L2 | 5 | 1 |
| b) What are honeypots and honeynets? Give examples. Explain the role of honeypots in detecting and analyzing attacks. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
